

G D HRUTHIK

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years of experience in designing and deploying end-to-end ML pipelines and computer vision workflows. Specialized in deep learning and scalable MLOps solutions, with expertise in ETL processes, SQL, and containerization. Recognized for transitioning models from research to production while delivering measurable operational improvements. Adept at collaborating with cross-functional teams to enhance data quality and drive innovative, real-world applications.

EDUCATION

New England College, Henniker, NH

Jun 2025

Master of Science, Data Science and Analytics

- **GPA:** 3.74

PROFESSIONAL EXPERIENCE

Sanofi

Apr 2024 - Present

Machine Learning Engineer

Boston, MA

- Collaborated with clinical teams to design efficient ETL processes and clean genomic and clinical trial datasets, enhancing experiment reproducibility.
- Containerized models with Docker and Kubernetes, enabling scalable deployment across oncology and immunology studies.
- Implemented model monitoring, versioning, and compliance checks according to FDA and GxP standards.
- Developed internal AI tools using Streamlit and FastAPI, increasing adoption of AI solutions among scientists by 40%.
- Integrated EHR and biomarker data into production pipelines with robust SQL queries, supporting translational medicine teams in biomarker discovery.
- Engineered a computer vision workflow using CNNs and PyTorch for microscopy image analysis, reducing protein misfolding detection time by 50% compared to manual review.

Cognizant Technology Services

Aug 2021 - May 2023

Machine Learning Engineer

Coimbatore, TN, India

- Engineered real-time anomaly detection models with Autoencoders and Isolation Forests, cutting equipment downtime by 20% for industrial clients.
- Constructed predictive maintenance pipelines using Random Forest algorithms, Kafka, and Azure Stream Analytics to accurately forecast equipment failures.
- Deployed a fraud detection solution on Azure ML, achieving inference speeds under 120ms and enhancing detection accuracy.
- Enhanced an e-commerce recommendation system with TensorFlow, leading to a 15% increase in click-through rates during A/B testing.
- Collaborated on end-to-end ML operations pipelines by standardizing containerization and monitoring processes, facilitating seamless model productization with cross-functional teams.

TECHNICAL SKILLS

- **Programming & Data Analysis:** Python, C++, R, SQL, Scala, Pandas, NumPy
- **Machine Learning:** Regression, Classification, Clustering, Dimensionality Reduction, Ensemble Methods (Bagging, Boosting, Stacking), Hyperparameter Tuning (Optuna, Grid Search)
- **Deep Learning & Computer Vision:** CNN, RNN, LSTM, BERT, CodonBert, ProGen2 Transformers, Attention Mechanisms, Reinforcement Learning, CLIP, Image Classification, Object Detection, Fine-tuning Models, Diffusion Models
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, DeepSpeed, Flask, Django
- **Natural Language Processing:** Text Preprocessing, Tokenization, Word Embeddings, Sentiment Analysis, LangChain, RAG, Hugging Face Transformers, OpenAI API
- **Generative AI:** GPT-based LLMs, Diffusion Models, Prompt Engineering
- **Vector Search & Retrieval:** FAISS, Pinecone, Chroma, Weaviate
- **MLOps & Model Deployment:** MLflow, DVC, Docker, Kubernetes, Azure Kubernetes Service (AKS), Streamlit, Ray, TorchServe, FastAPI, Triton Inference Server, TensorFlow Serving
- **CI/CD & Workflow Orchestration:** Jenkins, GitHub Actions, Apache Airflow, Azure DevOps, Luigi
- **Cloud Platforms:** AWS SageMaker, GCP Vertex AI, Azure ML Studio
- **Visualization & Reporting:** Tableau, Matplotlib, Seaborn
- **Data Engineering:** Hadoop, PySpark, Azure Databricks, Data Factory, Synapse Analytics, Snowflake
- **Project & Collaboration Tools:** Jupyter, Google Colab, VS Code, Agile, A/B Testing, Requirement Gathering